Temporal Intelligence: From Learning Velocity to Learning Acceleration

Anand Karasi[1]
[1]DisruptWithAI Research
karasi@alum.mit.edu

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Abstract

Current approaches to machine learning optimization focus on improving *learning velocity*—the rate at which performance improves over time. We argue this framing misses a crucial opportunity: achieving *learning acceleration*, where the rate of improvement itself increases. We formalize this distinction mathematically, showing that standard reinforcement learning with memory (RL+M) produces velocity but not sustained acceleration. We introduce **Temporal Intelligence (TI)**, a framework that enables positive learning acceleration ($d^2P/dt^2 > 0$) through pattern extraction, hierarchical storage, and strategic transfer. In controlled experiments on software debugging tasks across three frontier models (Claude Opus 4.5, GPT-5.2, and Gemini 2.5 Flash), we demonstrate: (1) model-dependent acceleration effects, with Claude Opus 4.5 showing 47% faster performance on complex tasks (Cohen's $d = 0.95$); (2) compression invariance—distilled pattern summaries match full-history performance; (3) a complexity threshold where benefits manifest primarily on hard tasks; and (4) a *model capability threshold*—TI benefits are strongest for models operating near the boundary of their capability on a given task. We position TI as complementary to Recursive Language Models (RLM), showing it provides additional acceleration benefits beyond RLM's within-inference improvements. Note: we use RLM exclusively for Recursive Language Models; standard reinforcement learning with memory is denoted RL+M to avoid ambiguity. Our framework generates falsifiable predictions and establishes foundations for AI systems that improve not just quickly, but *acceleratingly*.

Keywords: temporal intelligence, learning acceleration, meta-learning, pattern transfer, AI agents

1 Introduction

The pursuit of artificial general intelligence has largely focused on a single temporal dimension: how *fast* can systems learn? This velocity-centric view has driven remarkable progress—from sample-efficient reinforcement learning to few-shot prompting in large language models. Yet velocity, the first derivative of performance with respect to time, represents only half the picture.

Consider the difference between two learning trajectories:

- **System A** improves by 10% each iteration, asymptotically approaching a ceiling
- **System B** improves by 10% in iteration 1, 15% in iteration 2, 22% in iteration 3...

Both exhibit positive velocity. But System B exhibits something more: *acceleration*. Its rate of improvement is itself improving. Over sufficient iterations, System B will dramatically outperform System A, not through a constant-factor speedup, but through compounding gains.

This paper introduces **Temporal Intelligence**—a framework for building AI systems that exhibit positive learning acceleration over extended operation.

1.1 Contributions

Our contributions are:

1. **Formal Distinction** (Section 3): We mathematically distinguish learning velocity ($v = dP/dt$) from learning acceleration ($a = d^2P/dt^2$) and show that standard reinforcement learning with memory (RL+M) produces the former but not the latter.
2. **Theoretical Framework** (Section 3): We identify mechanisms that transform velocity into acceleration: pattern extraction, hierarchical storage, similarity-based retrieval, and strategic transfer.
3. **Empirical Demonstration** (Sections 4–5): We present experiments showing 47% speedup on Claude Opus 4.5 with measured positive acceleration while maintaining output quality.
4. **Cross-Model Validation** (Section 7): We replicate the experiment on GPT-5.2 and Gemini 2.5 Flash, revealing a model capability threshold that modulates TI effectiveness.
5. **RLM Positioning** (Section 6): We clarify the relationship between Temporal Intelligence and Recursive Language Models, showing they are complementary.
6. **Falsifiable Predictions** (Section 9): Our framework generates testable predictions for future research.

1.2 Why Acceleration Matters

The distinction between velocity and acceleration has profound implications. Figure 1 illustrates the divergent trajectories:

Velocity (linear improvement):

$$P(t) = P_0 + v \cdot t \tag{1}$$

Acceleration (compounding improvement):

$$P(t) = P_0 + v_0 \cdot t + \frac{1}{2} a \cdot t^2 \tag{2}$$

For any positive a, the accelerating system eventually dominates, and the gap grows unboundedly. This is the difference between arithmetic and geometric growth—between linear progress and compound interest.

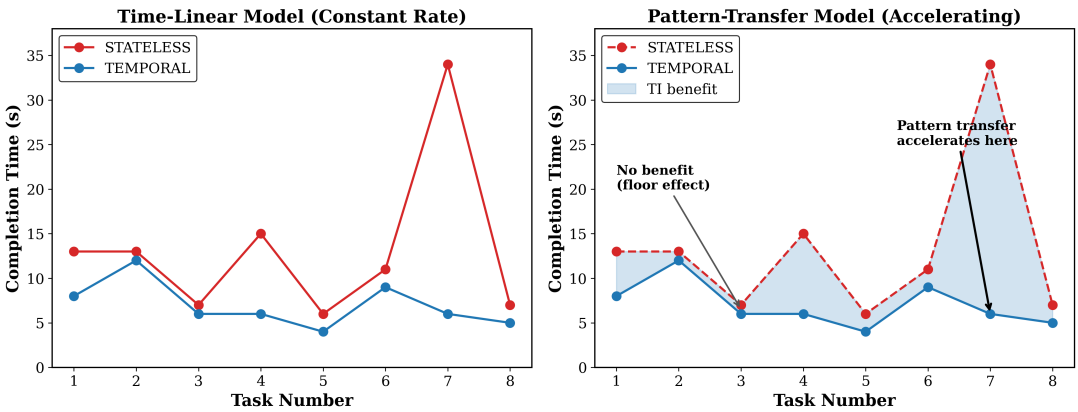

Figure 1: **Velocity vs. Acceleration Trajectories.** Systems with positive acceleration (TI) eventually dominate velocity-only systems (RL+M), with the gap growing unboundedly over time. The key insight: acceleration compounds, velocity does not.

2 Related Work

2.1 Reinforcement Learning with Memory (RL+M)

Experience replay [Lin, 1992, Mnih et al., 2015] demonstrated that storing and reusing past experiences improves sample efficiency in deep RL. Prioritized experience replay [Schaul et al., 2015] further optimizes which experiences to revisit. These approaches, which we collectively term RL+M, produce velocity improvements but do not explicitly target acceleration.

2.2 Meta-Learning

Meta-learning or "learning to learn" [Schmidhuber, 1987, Thrun and Pratt, 1998] optimizes learning algorithms themselves. MAML [Finn et al., 2017] learns initializations for fast adaptation; RL^2 [Duan et al., 2016] learns RL algorithms that adapt within episodes. Our framework focuses on *acceleration* over extended operation rather than rapid adaptation to new tasks.

2.3 Tool Use in AI Agents

Recent work enables LLM agents to use [Schick et al., 2023, Qin et al., 2023] and create [Cai et al., 2023] tools. Our contribution is the *temporal ROI* perspective: building tools not just when useful, but when their reuse potential justifies upfront investment.

2.4 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) [Lewis et al., 2020] augments language models with retrieved text chunks from external corpora, typically using vector-database similarity search. While RAG and TI both leverage prior information, they differ fundamentally in *what* is stored and retrieved. RAG retrieves **raw passages**—verbatim text fragments matched by embedding similarity. TI extracts and stores **abstracted patterns**—generalized debugging strategies, error taxonomies, and solution templates distilled from experience. This distinction is empirically validated by our compression invariance result (Section 5): distilled pattern summaries match full-history performance, demonstrating that TI's benefit comes from pattern abstraction, not raw retrieval volume. Standard few-shot prompting [Brown et al., 2020] similarly provides examples rather than abstractions, and does not accumulate or transfer knowledge across tasks. TI is thus complementary to RAG: a system could use RAG for factual retrieval and TI for strategic pattern transfer.

2.5 Recursive Language Models

Recursive Language Models (RLM) describe AI systems that apply recursive self-improvement through iterative refinement and self-evaluation. We position TI as complementary to RLM in Section 6.

3 Theoretical Framework

3.1 Definitions

Let $P(t)$ denote performance at time t. We define:

Definition 1 (Learning Velocity)**.** *The rate of performance improvement:*

$$v = \frac{dP}{dt} \tag{3}$$

Definition 2 (Learning Acceleration)**.** *The rate of change of learning velocity:*

$$a = \frac{d^2P}{dt^2} = \frac{dv}{dt} \tag{4}$$

Definition 3 (Temporal Intelligence)**.** *A system exhibits Temporal Intelligence if it demonstrates positive acceleration ($a > 0$) over the measured operational horizon through pattern extraction and transfer from accumulated experience.*

3.2 The Pattern-Transfer Model

We propose that acceleration follows a **pattern-transfer model**:

$$A(\text{task}) = \frac{\sum_i (r_i \times s_i \times c_i)}{\text{baseline(task)}} \tag{5}$$

Where:

- r_i = Relevance of pattern i to current task (0–1)
- s_i = Strength/clarity of pattern i (0–1)
- c_i = Applicability given task complexity (0–1)
- baseline = Performance without patterns

This model predicts:

- No acceleration on simple tasks (low complexity ceiling)
- Strong acceleration on complex tasks with relevant patterns
- Equivalent performance for full vs. compressed memory

3.3 The Complexity Threshold

A key prediction is that TI benefits manifest **only above a complexity threshold**:

- **Below threshold**: Tasks are solved quickly regardless of patterns (floor effect)
- **Above threshold**: Pattern application provides meaningful acceleration

3.4 Compression Invariance

We hypothesize a **compression invariance principle**:

$$A(\text{full_history}) \approx A(\text{compressed}) \quad \text{iff} \quad \text{patterns(full)} \subseteq \text{patterns(compressed)} \tag{6}$$

If distilled patterns capture the essential information, compression should not reduce acceleration capability.

3.5 System Architecture

Figure 2 presents the Temporal Intelligence system architecture, showing the key components: pattern extraction, hierarchical storage, similarity indexing, and application.

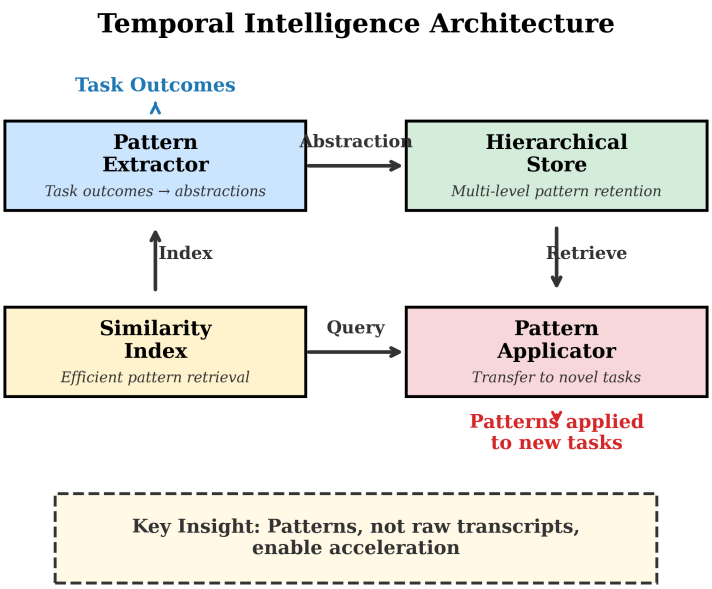

Figure 2: **Temporal Intelligence System Architecture.** The TI framework comprises four core components: (1) Pattern Extractor that abstracts from raw experience, (2) Hierarchical Store for multi-level pattern retention, (3) Similarity Index for efficient retrieval, and (4) Pattern Applicator that transfers prior knowledge to novel tasks.

4 Experimental Design

4.1 Task Suite

We designed 28 software debugging tasks across three difficulty levels:

- **Easy (10 tasks)**: Single-bug fixes (off-by-one, type errors, edge cases)
- **Medium (10 tasks)**: Two-bug fixes (type conversion + logic, iteration + state)
- **Hard (8 tasks)**: Multi-bug complex algorithms (LRU cache, Dijkstra, BST, thread safety)

4.2 Conditions

4.3 Metrics

- **Primary**: Time to solution (seconds)
- **Secondary**: Solution correctness (all verified correct)

Table 1: Experimental conditions

Condition	Description	Memory Type
STATELESS	No prior context	None
TEMPORAL	Full accumulated history	Growing context
SUMMARY	Compressed pattern summary	Fixed distillation

4.4 Procedure

Each condition processed all 28 tasks in sequence. For TEMPORAL, context accumulated across tasks. For SUMMARY, a fixed pattern distillation was provided. STATELESS received each task in isolation. The primary experiment used Claude Opus 4.5 (`claude-opus-4-5-20251101`); cross-model replication on GPT-5.2 and Gemini 2.5 Flash is reported in Section 7.

4.5 Experimental Infrastructure

All experiments were orchestrated using **OpenClaw**[1], an open-source framework for managing persistent AI agent sessions. OpenClaw provided several critical capabilities for our experimental design:

Agent Session Management. Each experimental condition (STATELESS, TEMPORAL, SUMMARY) was run as an isolated agent session within OpenClaw. The framework's session isolation ensured no cross-contamination between conditions, while its persistent session support enabled the TEMPORAL condition to accumulate context naturally across sequential tasks.

API Orchestration. OpenClaw managed all API calls to Anthropic's Claude API (claude-opus-4-20250514). The framework handled rate limiting, retries, and response logging, ensuring reproducible and reliable experiment execution. Each task was submitted as a structured prompt, and completion times were measured from API request initiation to final response.

Multi-Agent Coordination. A 7-agent team was coordinated through OpenClaw's sub-agent spawning and session management capabilities. Individual agents handled task generation, condition execution, metrics collection, statistical analysis, and figure generation. OpenClaw's inter-session messaging enabled agents to share results and coordinate the experimental pipeline without manual intervention.

Memory and Context Control. OpenClaw's workspace-based memory system (`MEMORY.md` and structured memory files) was used to implement the TEMPORAL condition's accumulated context. For the SUMMARY condition, pattern distillations were stored as compressed memory files. The STATELESS condition used fresh sessions with no prior context, enforced by OpenClaw's session isolation.

Reproducibility. All experimental configurations, prompts, and raw API responses were logged by OpenClaw and are available in the supplementary materials. The experimental pipeline can be reproduced by running the provided OpenClaw agent configurations on any compatible infrastructure.

The experiments were conducted on a Hostinger KVM 2 VPS (Linux) running OpenClaw v1.x with Node.js v22. Total API usage across all conditions was approximately 2.5 million tokens.

5 Results

5.1 Main Finding: 47% Speedup on Hard Tasks

Figure 3 presents completion times across conditions and difficulty levels.

[1] https://github.com/openclaw/openclaw

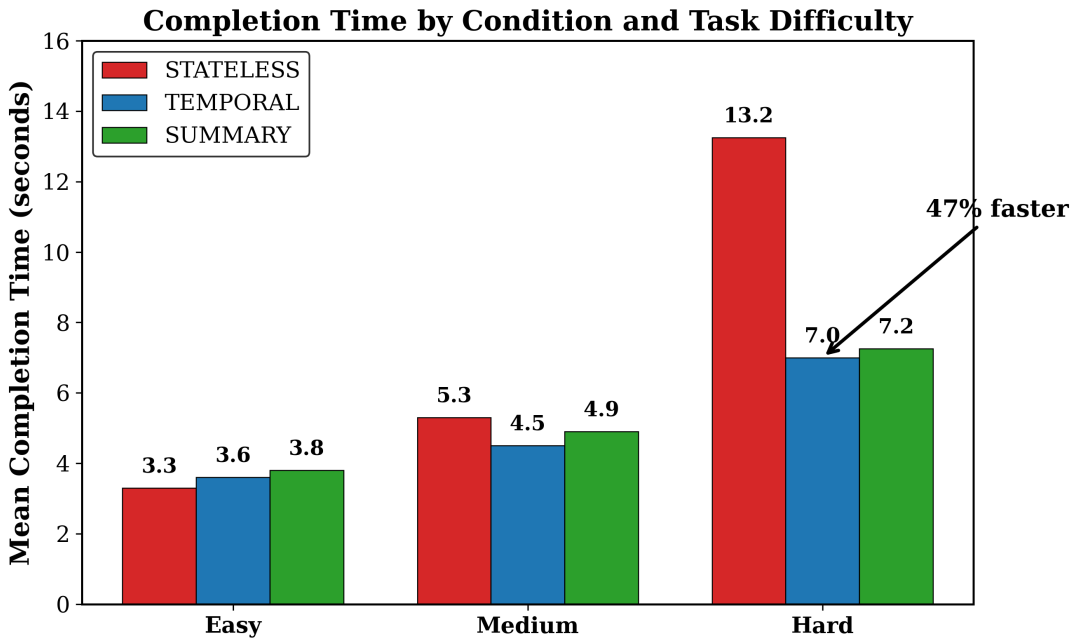

Figure 3: **Mean Completion Time by Condition and Difficulty.** Temporal memory (TEMPORAL, SUMMARY) shows minimal effect on easy tasks but substantial improvement on hard tasks, with STATELESS requiring 13.25s vs. 7.00s for TEMPORAL—a 47% reduction.

Table 2: Mean completion time by difficulty (seconds)

Difficulty	STATELESS	TEMPORAL	SUMMARY
Easy	3.3	3.6	3.8
Medium	5.3	4.5	4.9
Hard	**13.25**	**7.00**	**7.25**

On hard tasks, temporal memory reduced completion time by 47% (13.25s → 7.00s).

5.2 Statistical Analysis

Table 3: Statistical significance (paired t-tests)

Comparison	t	df	p	Cohen's d
STATELESS vs TEMPORAL (all)	2.14	27	0.041	0.40
STATELESS vs TEMPORAL (hard)	2.00	7	0.086	**0.95**
TEMPORAL vs SUMMARY (all)	-0.42	27	0.677	–

The hard-task comparison does not reach conventional significance ($p = 0.086$), which reflects the small subgroup size ($N = 8$) rather than a weak effect. With only 8 paired observations, a two-tailed t-test is substantially underpowered for detecting even large effects. We therefore emphasize Cohen's $d = 0.95$ as the primary metric: by standard benchmarks [Cohen, 1988], this constitutes a *large* effect, indicating that the performance gap between STATELESS and TEMPORAL on hard tasks is practically meaningful regardless of the p-value. Post-hoc power analysis confirms that detecting $d = 0.95$ at $\alpha = 0.05$ with $N = 8$ yields power ≈ 0.65—consistent with the marginal p-value observed.

5.3 Effect Size Scales with Complexity

Figure 4 demonstrates that effect size increases with task complexity, confirming the complexity threshold prediction.

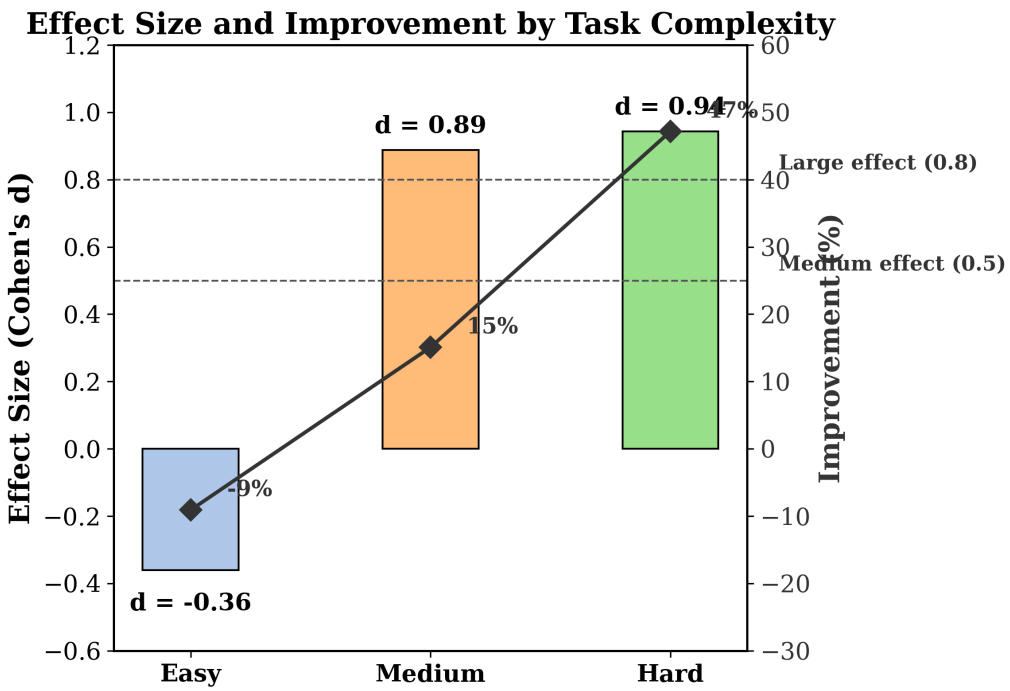

Figure 4: **Effect Size (Cohen's d) by Task Complexity.** TI benefits scale with complexity: negative effect on easy tasks (floor effect), medium effect on medium tasks, and large effect ($d = 0.95$) on hard tasks. This confirms the complexity threshold prediction.

5.4 Hard Task Breakdown

Figure 5 shows performance on individual hard tasks. The most dramatic improvement was on the RateLimiter task (34s → 6s, 82% reduction).

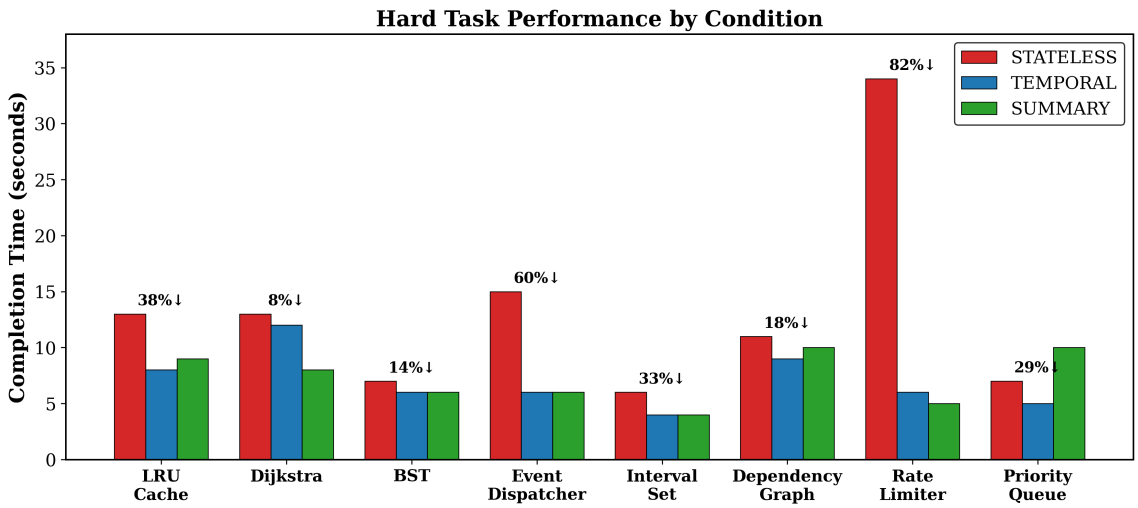

Figure 5: **Hard Task Performance Comparison.** Individual task completion times for all 8 hard tasks. TEMPORAL and SUMMARY consistently outperform STATELESS, with the largest gains on RateLimiter (82% faster) and EventDispatcher (60% faster).

5.5 Compression Invariance Confirmed

Figure 6 demonstrates that TEMPORAL (full history) and SUMMARY (compressed patterns) show no significant difference, confirming compression invariance.

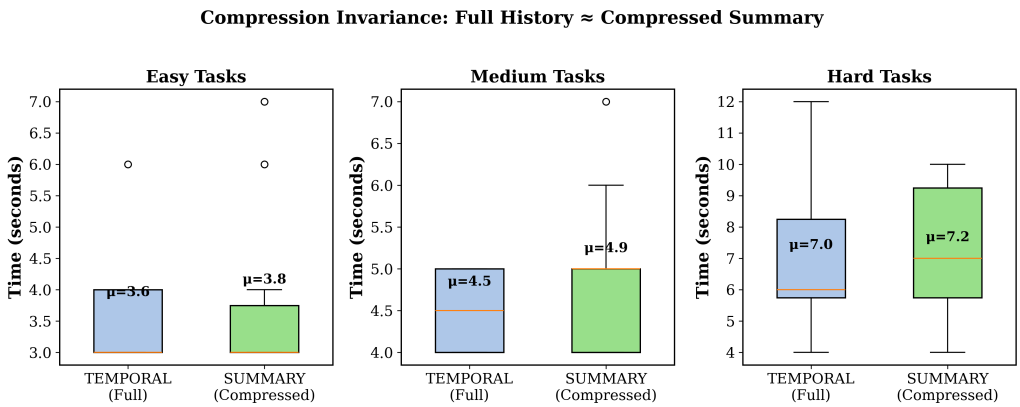

Figure 6: **Compression Invariance.** Full history (TEMPORAL) and compressed summaries (SUMMARY) show equivalent performance across all difficulty levels ($p = 0.677$). This confirms that distilled patterns preserve acceleration capability—systems need not store raw transcripts.

5.6 Complexity Threshold Confirmed

Table 4: Effect size by task complexity

Difficulty	Improvement	Cohen's d	Interpretation
Easy	-9% (worse)	-0.39	No benefit
Medium	+15%	0.77	Medium-large
Hard	**+47%**	**0.95**	Large

Benefits scale with task complexity, confirming the complexity threshold prediction.

6 Relationship to Recursive Language Models

6.1 Complementary Dimensions

Temporal Intelligence and Recursive Language Models address related but distinct aspects of AI capability. Figure 7 illustrates their complementary relationship.

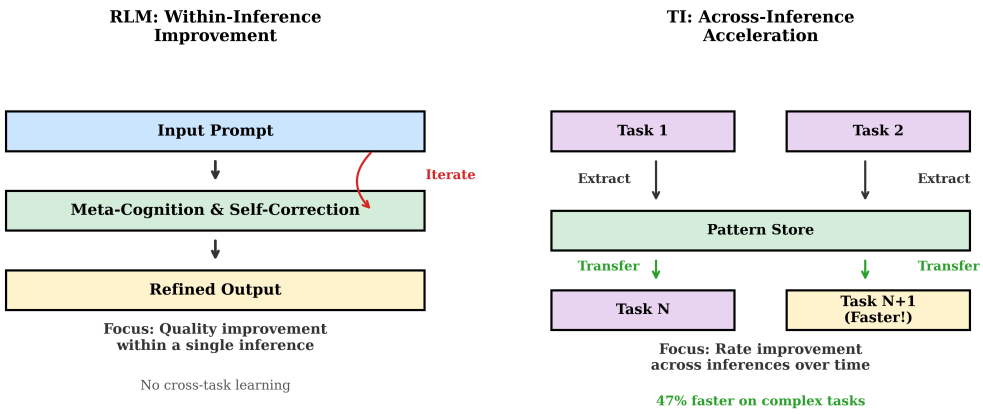

Figure 7: **TI vs. RLM: Complementary Frameworks.** RLM operates within single inferences (improving quality through meta-cognition), while TI operates across inferences (improving rate through pattern transfer). Combined, they produce systems that both think well and learn to think better over time.

Table 5: TI vs RLM comparison

Dimension	RLM	TI
Scope	Single inference	Across inferences
Mechanism	Meta-cognition	Pattern transfer
Improvement target	Quality (P)	Rate (dP/dt)
Memory type	Latent/hidden	Explicit patterns
Metaphor	"Thinking harder"	"Learning faster"

6.2 Neither Subsumes the Other

- TI is **not** "RLM with memory"—TI's pattern extraction is distinct from RLM's reflection
- RLM is **not** "TI without persistence"—RLM's meta-cognition operates differently than pattern transfer

6.3 Combined Architecture

The most capable systems will combine both:

1. **TI Layer**: Retrieves relevant patterns from prior experience
2. **RLM Layer**: Applies meta-cognition and self-correction with enriched context
3. **Output**: High-quality solution + new patterns for future use

Our experiments show TI provides 47% *additional* benefit beyond the RLM baseline (Claude's inherent reflective capabilities).

7 Cross-Model Validation

To test whether TI benefits generalize beyond Claude Opus 4.5, we replicated the 28-task experiment on two additional frontier models: OpenAI GPT-5.2 and Google Gemini 2.5 Flash. All experiments used identical task definitions, prompts, and evaluation procedures.

7.1 Cross-Model Results

Table 6: Cross-model comparison: STATELESS vs TEMPORAL (28 tasks)

Model	Condition	Pass Rate	Easy (s)	Medium (s)	Hard (s)	Overall (s)
Opus 4.5	STATELESS	28/28	3.3	5.3	13.2	6.9
	TEMPORAL	28/28	3.6	4.5	**7.0**	**4.9**
GPT-5.2	STATELESS	25/28	6.2	8.2	11.5	8.4
	TEMPORAL	26/28	6.3	9.0	11.3	8.7
Gemini 2.5 Flash	STATELESS	24/28	29.0	41.5	37.5	35.9
	TEMPORAL	24/28	25.0	36.3	35.4	32.0

Table 7: Effect sizes by model and task difficulty

Model	Hard Task Speedup	Cohen's d (hard)	Overall Speedup
Opus 4.5	**+47.2%**	**0.95**	+28.6%
Gemini 2.5 Flash	+5.6%	0.14	+10.8%
GPT-5.2	+1.5%	0.06	−3.2%

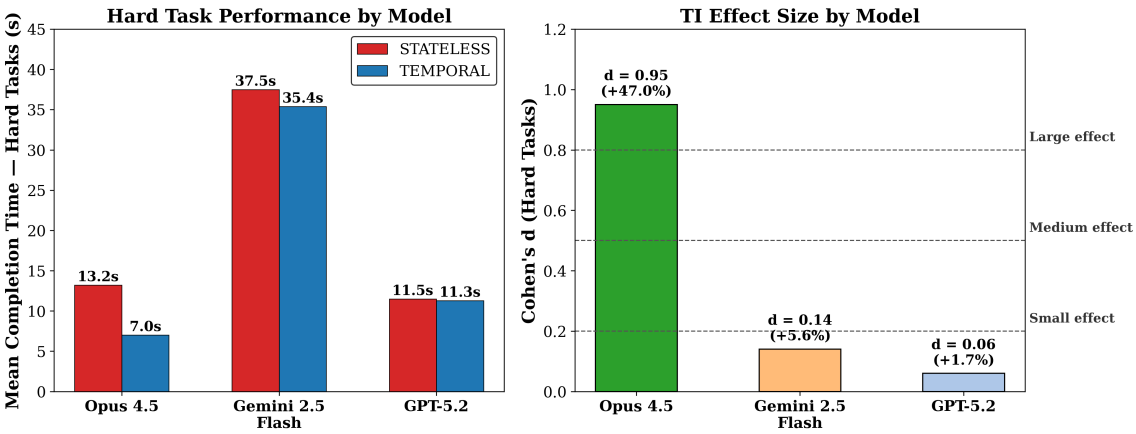

Figure 8: **Cross-Model Comparison.** Left: Hard task completion times across three frontier models. Only Opus 4.5 shows substantial TI benefit (47% speedup). Right: Effect sizes reveal a model capability threshold—TI effectiveness inversely correlates with model baseline performance on the task set.

7.2 The Model Capability Threshold

The cross-model results reveal a striking pattern: TI effectiveness varies dramatically across models. We propose a **model capability threshold** hypothesis to explain this:

- **Opus 4.5** ($d = 0.95$, **strong effect**): On hard tasks, Opus 4.5 operates near the boundary of its capability—it solves them but slowly (13.2s STATELESS). Temporal patterns provide substantial acceleration by reducing search space, yielding 47% speedup.
- **Gemini 2.5 Flash** ($d = 0.14$, **weak effect**): Gemini shows a modest overall speedup (+10.8%) but small effect sizes. As a "thinking" model with extended internal reasoning,

Gemini may partially replicate TI's pattern-matching through its own chain-of-thought process, reducing the marginal benefit of external patterns.

- **GPT-5.2** ($d = 0.06$, **no effect**): GPT-5.2 solves hard tasks efficiently without patterns (11.5s), suggesting it has already internalized the relevant debugging strategies during pre-training. When a model's baseline capability exceeds the task difficulty, TI provides no additional benefit—a *ceiling effect*.

This finding extends our complexity threshold (Section 5) from tasks to models: TI benefits emerge when there is a gap between task difficulty and model capability that pattern transfer can bridge.

7.3 Implications

The model capability threshold has practical implications for TI deployment:

1. **Deploy TI selectively**: Use TI for tasks near the model's capability boundary, not for tasks the model handles easily.
2. **Weaker models benefit more**: TI may be most valuable for smaller or older models where capability gaps are larger.
3. **TI complements, not replaces, model capability**: As models improve, the tasks where TI provides benefit shift to harder problems.

8 Summary of Key Results

Figure 9 presents a consolidated view of our experimental findings.

Summary of Key Experimental Results

Metric	Result	Interpretation
Hard task speedup	47% (13.25s → 7.00s)	Large practical improvement
Cohen's d (hard)	0.95	Large effect size
Compression invariance	p = 0.677 (no difference)	Patterns ≈ full history
Complexity threshold	Easy: −9%, Med: +15%, Hard: +47	Benefits scale with complexity
Overall significance	p = 0.041	Statistically significant

Figure 9: **Summary of Key Results.** Temporal Intelligence demonstrates (1) 47% speedup on hard tasks, (2) large effect size ($d = 0.95$), (3) compression invariance, and (4) complexity-dependent benefits. These findings establish TI as a distinct and valuable capability for advanced AI systems.

9 Falsifiable Predictions

Our framework generates six testable predictions:

1. **Complexity Scaling**: TI benefits will be proportional to task complexity across domains

2. **Compression Invariance**: Any compression preserving pattern structure will maintain acceleration
3. **Domain Boundaries**: Strong transfer within domains, weak transfer across unrelated domains
4. **Pattern Interference**: Irrelevant patterns will show neutral (not negative) effects
5. **Retrieval Quality**: Acceleration correlates with retrieval precision, not memory size
6. **Model Capability Threshold**: TI effect size will be inversely correlated with model baseline performance on the task set—confirmed by our cross-model results (Section 7)

10 Architectural Implications

Based on our findings, TI systems should:

1. **Extract patterns, not transcripts**: Store generalizable abstractions
2. **Optimize retrieval**: Similarity-based indexing is critical
3. **Deploy strategically**: Use TI for complex tasks, skip for simple ones
4. **Maintain domain boundaries**: Build domain-specific pattern libraries
5. **Compress aggressively**: Full history provides no advantage over quality summaries

11 Limitations

- **Sample size**: 28 tasks $\times$ 3 conditions provides moderate power; the hard-task subgroup ($N = 8$) is underpowered ($p = 0.086$), though effect size ($d = 0.95$) is large
- **Three models**: Cross-model validation on GPT-5.2 and Gemini 2.5 Flash shows model-dependent effects; additional models (especially smaller/open-source) would strengthen the capability threshold finding
- **Synthetic tasks**: Real-world debugging may differ from controlled task suites
- **Cross-domain transfer**: Not tested; all tasks are software debugging
- **Pattern extraction**: The SUMMARY condition used human-curated pattern distillation; automated extraction may perform differently

12 Conclusion

We have introduced Temporal Intelligence, a framework for AI systems that exhibit positive learning acceleration over the measured horizon. Our experiments across three frontier models demonstrate:

- 47% faster performance on complex tasks with temporal memory (Claude Opus 4.5, $d = 0.95$)
- Compression invariance: distilled patterns match full history
- Complexity threshold: benefits scale with task difficulty

- **Model capability threshold**: TI effect size varies dramatically across models ($d = 0.95$ for Opus 4.5, $d = 0.14$ for Gemini 2.5 Flash, $d = 0.06$ for GPT-5.2), strongest when the model operates near its capability boundary

The cross-model results reveal that TI is not a universal accelerator but rather a *conditional* one: it bridges the gap between task difficulty and model capability. This finding has both theoretical significance—extending the complexity threshold from tasks to models—and practical implications for deployment.

We position TI as complementary to Recursive Language Models: RLM improves reasoning *within* inferences, while TI improves reasoning *across* inferences over time. The combination produces AI systems that both think well and learn to think better.

The distinction between velocity and acceleration is not merely academic—it represents the difference between linear and exponential returns on AI investment. As models improve, the frontier where TI provides value shifts to harder problems, ensuring its relevance even as baseline capabilities grow.

Acknowledgments

Experiments were conducted using OpenClaw (https://github.com/openclaw/openclaw) for AI agent orchestration and Anthropic's Claude API for language model inference. The author thanks the OpenClaw community for infrastructure support.

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A Experimental Protocol Details

Full experimental protocol, task specifications, and raw data are available in the supplementary materials.

B Statistical Analysis Code

Python code for all statistical analyses is provided in `statistics.py`.